Solutions to Assignment 2, ME647, 2025-26-II

Submitted by Shaurya Singh, 218070969

Problem 1: Kolmogorov Theory, Spectra and Turbulence Scaling Laws

You can work with the same data as the previous assignment: A two-dimensional cut from a direct numerical simulation of three-dimensional homogeneous, isotropic turbulence from the Johns Hopkins Turbulence Database. The details of your dataset are as follows, along with dimensionless parameters:

- **Grid Size:** $N_x \times N_y = 1024 \times 1024$ grid cells (which is a 2D plane, at $z = 0$, from a 1024^3 simulation). The lateral size $L_x = L_y = 2\pi$, and the length of each grid cell is $dx = L/N_x$. The kinematic viscosity $\nu = 0.000185$ and integral lengthscale $\mathcal{L} = 1.364$.
- **Velocity fields in dataset:** u, v, w or u_i, with $i \in \{1, 2, 3\}$
- **Boundaries:** Periodic (i.e. $u_i(l, y) = u_i(l + 2\pi, y)$)

Calculate, plot, and comment on the following measurements:

1. Verify Parseval's theorem using a line-cut of the data (say u along x at $y = 0$) and its Fourier transform. Explain the result. (5 Marks)
2. The energy spectra
$$E(k) = \frac{(|\hat{u}|^2 + |\hat{v}|^2)}{2},$$
where $\hat{u}$ is the Fourier Transform of the u velocity field and k is the wavenumber. Calculate the energy spectra along lines in the x-direction, and average over the y direction. Plot the averaged energy spectrum as a function of k (calculate the k values based upon the length of the data and its spatial resolution appropriately) on a **log-log scale**. Label and show the scaling exponent $k^{-5/3}$ in the inertial range with a line-fit.

   [Make sure you show the relevant axis-range! You will lose marks for blindly showing nonsensical axes ranges.]

   (8 Marks)
3. In the above example, the spectra is calculated in 1D, and hence k is simply a wavenumber. Now compute the two-dimensional energy spectrum using the full 2D velocity fields (u, v) to obtain $E(\mathbf{k})$ where $\mathbf{k} = k_x\hat{i} + k_y\hat{j}$. This is a spectral field in wavevector space. Perform a shell-averaging over wavenumber shells $k - 1/2 \leq k < k + 1/2$ for $k \in [k_{\min}, k_{\max}]$, to obtain the spherically averaged energy spectrum
$$E(k) = \sum_{k-1/2 \leq |\mathbf{k}| < k+1/2} E(\mathbf{k})$$

over the scalar wavenumber

$$k = \sqrt{k_x^2 + k_y^2}.$$

Again label and show the scaling exponent in the inertial range with a line-fit, and compare the spectra to the one in the previous problem. *[You must understand the concept and algorithm employed for this, as it may be evaluated later.]*

(10 Marks)

4. Plot the longitudinal and transverse velocity correlation functions from the velocity fields.

(12 Marks)

5. Longitudinal structure functions are defined as

$$S_p(r) = \langle \Delta u^p(r) \rangle = \left\langle \left| [\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})] \cdot \frac{\mathbf{r}}{|\mathbf{r}|} \right|^p \right\rangle \propto r^{\zeta_p}$$

where the angle brackets denote averaging over all orientations of $\mathbf{r}$ (for a fixed $|\mathbf{r}|$) and over many centres $\mathbf{x}$. For ease of coding, you can take $\mathbf{r}$ along the x-direction alone, or for more accuracy, take arbitrary orientations of $\mathbf{r}$ at each point $\mathbf{x}$. Keep in mind, and use the fact, that the data is periodic in space. Perform the averaging over many points!

(a) Plot $S_p(r)$ v/s r for $p \in \{1, 2, 3, 4, 5, 6, 7\}$ and $r \in [0, L/2]$ on a log-log scale. Plot these in the same figure, with clean legends and labels.

(10 Marks)

(b) Verify Kolmogorov's 4/5th law for $S_3(r)$.

(5 Marks)

(c) It is typically difficult to find the scaling (inertial) range to obtain the structure function scaling exponents ζ_p (it needs a lot of statistics). You can, however, plot $S_p(r)$ v/s $S_3(r)$ instead, on a log-log scale, which improves the scaling range significantly, i.e. you find the scaling of $S_p(r)$ versus $S_3(r)$, and not with r itself. This technique, used widely in turbulence study, is known as ESS: *Extended Self-Similarity* (note that $\zeta_3 = 1$, by default in ESS). Plot the ESS profiles of all the $S_p(r)$, on the same plot, using clear legends and labels.

(10 Marks)

(d) To the ESS plots above, add line fits in the scaling-range to obtain the scaling exponents ζ_p and errorbars for the different p-values. Then plot the scaling exponents ζ_p v/s p, with errorbars, against Kolmogorov's prediction of $\zeta_p = p/3$. What is the source of this anomalous scaling of exponents, and why does the deviation prominently increase at large values of p?

(10 Marks)

Solution

1. Parseval's theorem states that the total energy of a signal is conserved in both the physical space and the Fourier space. For a discrete signal $u(x_n)$ and its discrete Fourier transform $\hat{u}(k_m)$ this means that

$$\sum_{n=0}^{N-1} |u(x_n)|^2 = \frac{1}{N} \sum_{m=0}^{N-1} |\hat{u}(k_m)|^2 \tag{1}$$

For a line-cut of the velocity field ($u(x, y = 0)$), Parseval's theorem would be equivalent of saying that total kinetic energy computed in physical space (sum of squared velocities along the line) must be equal to the total energy in spectral space (sum of squared Fourier velocities).

Verification of this statement numerically with the following script:

```
u_line = u[:,0]
en = np.sum(np.abs(u_line) ** 2)

u_line_fourier = np.fft.fft(u_line)
frequency_en = np.sum(np.abs(u_line_fourier) ** 2)/nx

print(f"total energy in phsyical space= {en}")
print(f"total energy in spectral space = {frequency_en}")
```

```
total energy in phsyical space= 400.06731982553725
total energy in spectral space = 400.06731982553737
```

Assuming that the difference/error comes from floating-point round-off errors, the theorem holds. We can say that this transformation doesn't create a loss of information, and is an equivalent representation to study the the energy distribution of the flow.

2. To calculate the average energy spectrum (averaged over y), the script below was used. Note that for fitting of the scaling exponent, I basically performed hit and trial on the inertial subrange limits to get it close to -5/3 and verified it visually.

```

uhat = np.fft.fft(u, axis=0) #DFT in x axis only
vhat = np.fft.fft(v, axis=0)

k = 2 * np.pi * np.fft.fftfreq(nx, dx) # wavenumbers
E = (np.abs(uhat)**2 + np.abs(vhat)**2) / 2
Ek = np.mean(E, axis=1) #mean along y axis

positive_k = np.arange(0, nx//2 + 1)
k_plus = k[positive_k] # energy spectrum only cares about magnitude

Ek_plus = Ek[positive_k]
Ek_plus[1:-1] *= 2  # Double for k=1 to k=nx//2-1 to account for negative
   frequencies

fig, ax = plt.subplots(figsize=(20, 18))
ax.loglog(k_plus, Ek_plus, 'b-', linewidth=2, label='Energy Spectrum') #
   plot spectrum
```

```
mink = 0.01
maxk = 30 #inertial subrange comes from hit and trial
inertialsubrange = (k_plus > mink) & (k_plus < maxk)
k_inertial = k_plus[inertialsubrange]
Ek_inertial = Ek_plus[inertialsubrange] # for polyfit

logk = np.log(k_inertial)
logE = np.log(Ek_inertial)
coeffs = np.polyfit(logk, logE, 1) #fitting

k_fit = np.linspace(k_inertial[0], k_inertial[-1], 100)
E_fit = np.exp(coeffs[1] + coeffs[0] * np.log(k_fit))

ax.loglog(k_fit, E_fit, 'r--', linewidth=4, label=f'Fit: k^({coeffs[0]:.3f
   })') #plot fit

print(f"Inertial range: k = [{mink}, {maxk}]")
print(f"Fitted scaling exponent: {coeffs[0]}")
print(f"Expected scaling exponent: {-5/3}")
```

Output is as follows:

```
Inertial range: k = [0.01, 30]
Fitted scaling exponent: -1.7630
Expected scaling exponent: -1.6667
```

Plot with fit:

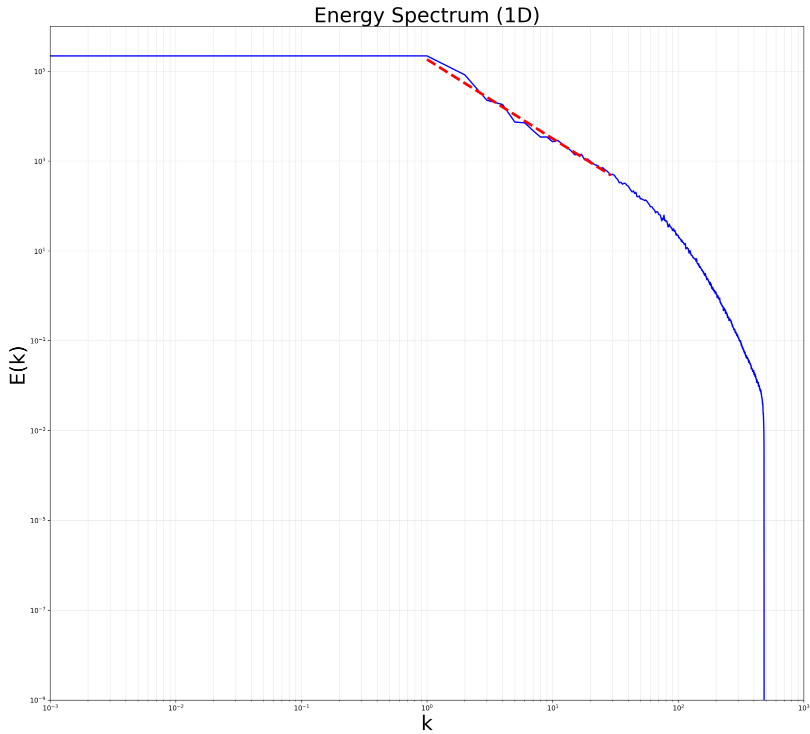

Figure 1: Log-log plot of the 1D energy spectrum $E(k)$ averaged over y, along with a fitted inertial range scaling showing approximate $k^{-5/3}$ behavior.

4. Script:

```
r_max = nx // 2
r_distances = np.arange(1, r_max) * dx

#Longitudinal correlation
R_L = np.zeros(len(r_distances))
for i, r_sep in enumerate(r_distances):
    r_int = int(np.round(r_sep / dx))
    du_prod = np.mean(u[r_int:, :] * u[:-r_int, :])
    R_L[i] = du_prod / np.mean(u**2)

#Transverse correlation
R_T = np.zeros(len(r_distances))
for i, r_sep in enumerate(r_distances):
    r_int = int(np.round(r_sep / dx))
    dv_prod = np.mean(v[r_int:, :] * v[:-r_int, :])
    R_T[i] = dv_prod / np.mean(v**2)
```

Plot:

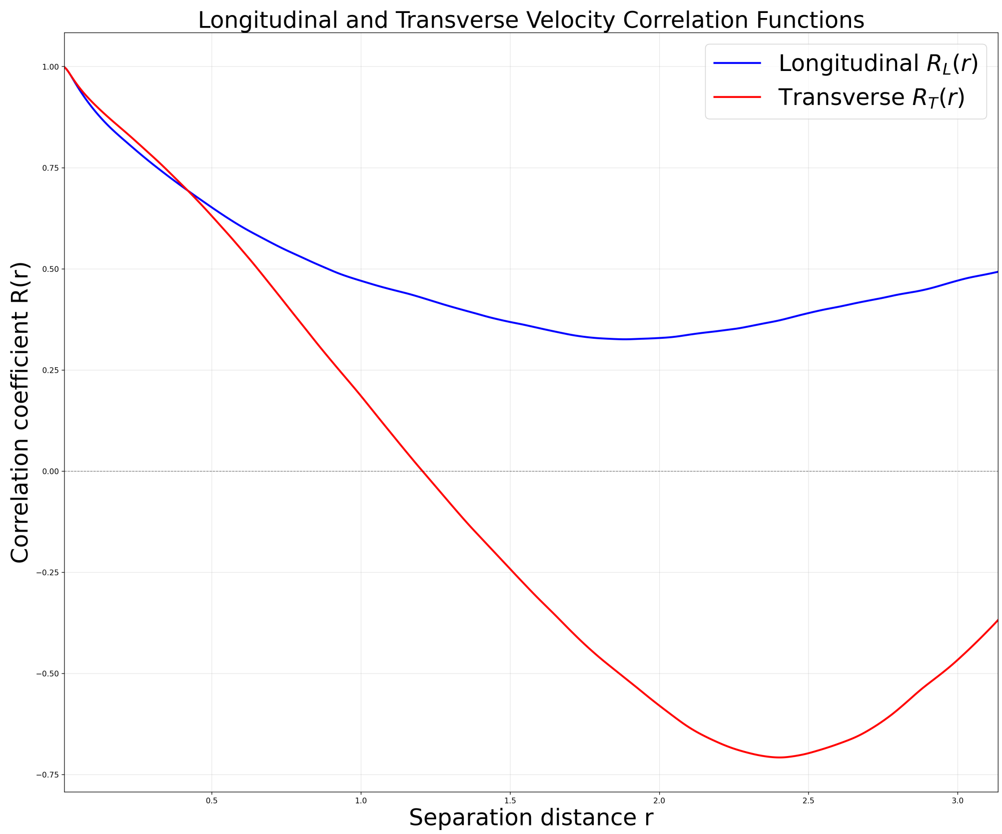

Figure 2: Longitudinal (R_L) and transverse (R_T) velocity correlation functions as a function of separation distance r.

5.a Script:

```
p_values = np.array([1, 2, 3, 4, 5, 6, 7])

r_max = int(L / (2 * dx))
r_array = np.arange(1, r_max) * dx

S_p = {}
for p in p_values:
    S_p[p] = np.zeros(len(r_array))

# Calculate structure functions
for i, r_sep in enumerate(r_array):
    r_int = int(np.round(r_sep / dx))
    # For each order p, calculate the p-th moment
    for p in p_values:
        sum_val = 0
        count = 0
        for j in range(len(u) - r_int):
            for k in range(len(u[0])):
                # Get velocity difference at this separation
                delta_u = u[j + r_int, k] - u[j, k]
                sum_val += np.abs(delta_u)**p
                count += 1
```

```
        S_p[p][i] = sum_val / count
```

Plot:

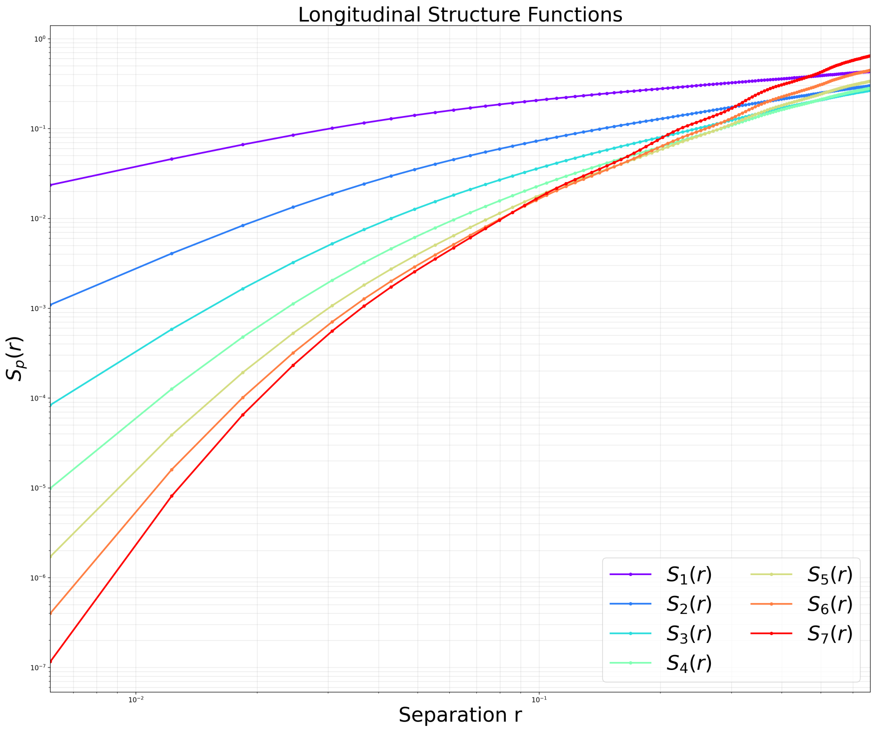

Figure 3: Log-log plot of longitudinal structure functions $S_p(r)$ for orders $p = 1$ to 7 as a function of separation distance r.

5.c Kolmogorov's 4/5th law:

$$S_3(r) = -\frac{4}{5}\epsilon r \tag{2}$$

Approximate dissipation rate was estimated using:

$$\epsilon \approx \frac{U^3}{L} \tag{3}$$

where $U = \sqrt{\langle u_i^2 \rangle}$ and $L = 1.346$. The computed $S_3(r)$ should follow the linear relationship $S_3(r) \propto -\epsilon r$ in the inertial range.

Obtained plot:

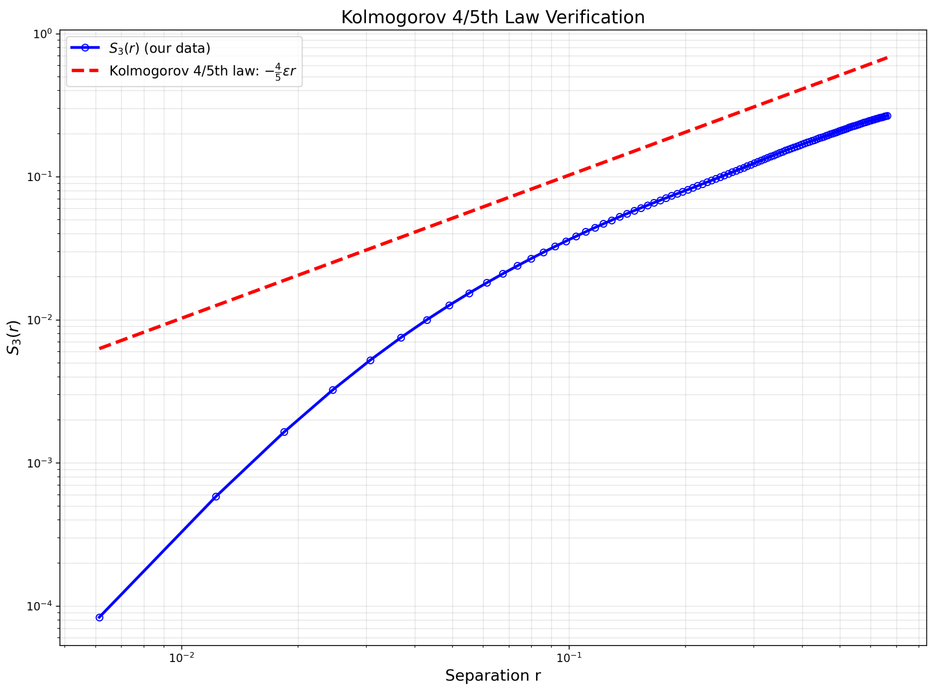

Figure 4: Verification of Kolmogorov's 4/5th law: third-order structure function $S_3(r)$ showing approximate linear scaling with r in the inertial range.

The measured $S_3(r)$ exhibits approximately linear behavior in logarithmic space across the inertial range

5.c Plot:

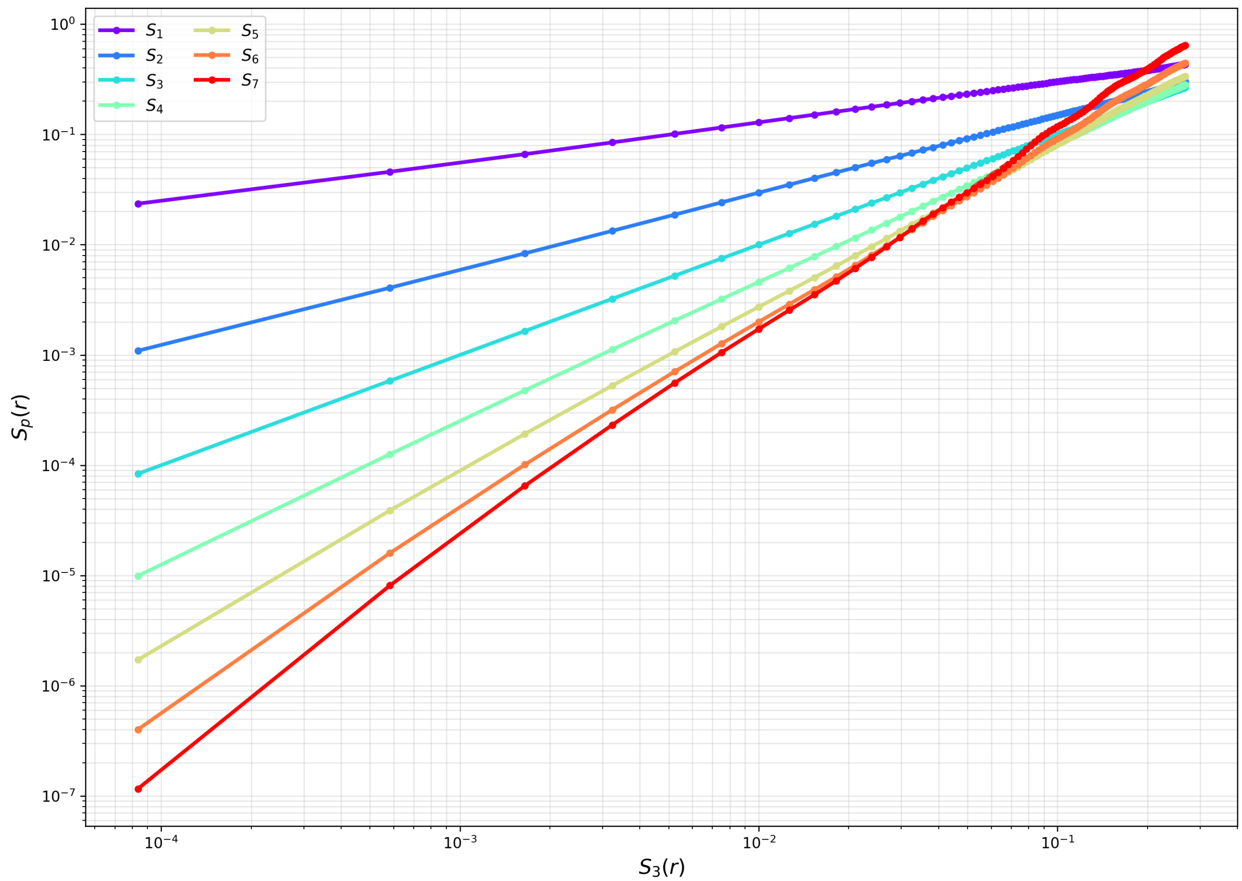

Figure 5: Extended Self-Similarity (ESS) plot: $S_p(r)$ versus $S_3(r)$ on a log-log scale for different orders p.

5.d Disclosure: AI help was used in writing the code for the plot. Combined script used for both parts (c and d):

```
zeta_p = np.zeros(len(p_values))
zeta_p_err = np.zeros(len(p_values))

fig, ax = plt.subplots(figsize=(22, 18))
for i, p in enumerate(p_values):
    # Plot S_p vs S_3
    ax.loglog(S_p[3], S_p[p], 'o-', color=colors[i], linewidth=2.5,
       markersize=4, label=f'$S_{{{p}}}$')

    log_S3 = np.log(S_p[3])
    log_Sp = np.log(S_p[p])

    n_data = len(log_S3)
    fit_idx = np.arange(int(0.2*n_data), int(0.8*n_data)) #(ignore edges)
```

```
    coeffs_ess = np.polyfit(log_S3[fit_idx], log_Sp[fit_idx], 1, cov=True)
    slope = coeffs_ess[0][0]
    intercept = coeffs_ess[0][1]
    cov_matrix = coeffs_ess[1]
    slope_err = np.sqrt(cov_matrix[0, 0])

    zeta_p[i] = slope
    zeta_p_err[i] = slope_err

    S3_fit = np.logspace(np.log10(S_p[3][fit_idx[0]]), np.log10(S_p[3][
       fit_idx[-1]]), 100)
    Sp_fit = np.exp(intercept + slope * np.log(S3_fit))
    ax.loglog(S3_fit, Sp_fit, '--', color=colors[i], linewidth=2, alpha
       =0.7)

ax.set_xlabel('$S_3(r)$', fontsize=30)
ax.set_ylabel('$S_p(r)$', fontsize=30)
ax.set_title('Extended Self-Similarity (ESS)', fontsize=30)
ax.grid(True, which='both', alpha=0.3)
ax.legend(fontsize=30, ncol=2)
plt.savefig('structure_functions_ess.png', bbox_inches='tight', dpi=240)
plt.show()

kolmogorov_pred = p_values / 3

fig, ax = plt.subplots(figsize=(12, 8))
ax.errorbar(p_values, zeta_p, yerr=zeta_p_err, fmt='bo-', linewidth=2.5,
   markersize=8,
            capsize=5, capthick=2, label='Our measurements',
               markerfacecolor='none')
ax.plot(p_values, kolmogorov_pred, 'r--', linewidth=2.5, label='Simple
   theory: $\\zeta_p = p/3$')
ax.set_xlabel('Order p', fontsize=14)
ax.set_ylabel('Scaling exponent $\\zeta_p$', fontsize=14)
ax.set_title('Anomalous Scaling', fontsize=16)
ax.grid(True, alpha=0.3)
ax.legend(fontsize=12)
plt.savefig('scaling_exponents_zeta_p.png', bbox_inches='tight', dpi=240)
plt.show()
```

Plot of scaling exponents with p:

The primary source of anomalous scaling is that Kolmogorov's theory assumes a homogeneous, self-similar cascade, but in reality, the flow is not self-similar.

At large values of p, the intermittencies become very large in value due to big exponents and start dominating the higher order moments, causing the deviation to increase.

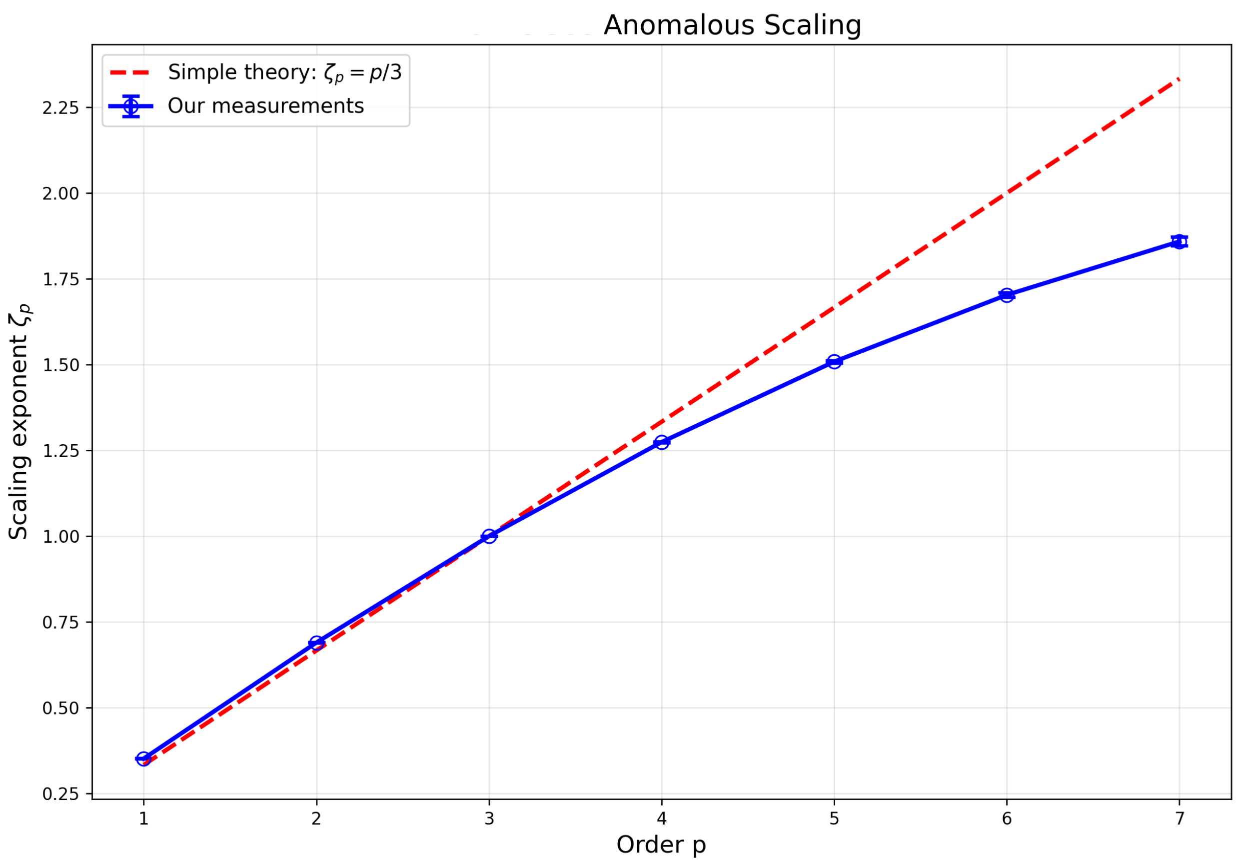

Figure 6: Scaling exponents ζ_p as a function of order p with error bars, compared against Kolmogorov's prediction $\zeta_p = p/3$, showing anomalous scaling.